Bank Customer Churn Analysis

Project summary: Power BI report on 10,000 retail bank customers in France, Germany and Spain

10,000	2,037	20.4%	24.3%
Total customers	Customers lost	Churn rate (target 15%)	Share of total balance held by churned customers

Executive summary

About one in five customers (2,037 of 10,000) has left the bank. That is a churn rate of 20.4%, which is 5.4 percentage points above the 15% target built into the report. Reaching the target would mean keeping about 537 more customers. Churned customers also held 24.3% of the total balance, so the loss is financial as well as a matter of headcount.

Churn is not spread evenly. It concentrates in a few clear segments: German customers, customers aged 41 to 60, customers who hold three or more products, and inactive members. Credit score and credit card ownership make almost no difference.

Project objective

Build an interactive Power BI report that shows who is leaving, which segments are most at risk, and where retention effort should go first.

Key findings at a glance

- **Germany:** 32.4% churn, about twice France (16.2%) and Spain (16.7%). Germany holds 25% of customers but 40% of all churn.
- **Age:** churn peaks at 56.2% for ages 51-60. Ages 41-60 are 31% of customers but 61% of churn.
- **Products:** customers with 3-4 products churn at 85.9% (280 of 326). Two products is the healthiest group at 7.6%.
- **Activity and gender:** inactive members churn at 26.9% (active 14.3%). Women churn at 25.1% (men 16.5%).
- **Balance:** churn rises with balance, from 13.8% at zero to 25.0% in the 100k-200k band.
- **Not drivers:** credit score bands all sit near 20%, and card ownership gives 20.2% vs 20.8%.

Data and model

The report is built on one customer table and four supporting tables. The supporting tables give ordered bands for charts, and a country table holds the flag image used in the slicer.

Table	Contents	Role
Customer Data	10,000 rows. Customer ID, country, gender, age, tenure, balance, credit score, products, credit card status, activity status, churn status, and target, min and max churn rate.	Main table
Age Groups	Seven bands: under 20, 21-30, 31-40, 41-50, 51-60, 61-70, over 71	Banding and sort order
Credit Scores	Six bands from 400 or below up to 801 and above	Banding and sort order
Acc Balance Groups	Balance bands from 0 up to over 200k	Banding and sort order
URL	Country name and a flag image link for France, Germany and Spain	Flag slicer

Relationships

Four many-to-one relationships, each filtering in a single direction: Customer Data to Age Groups (Age Group), to Credit Scores (Credit Scores), to Acc Balance Groups (Acc. Balance) and to URL (Country).

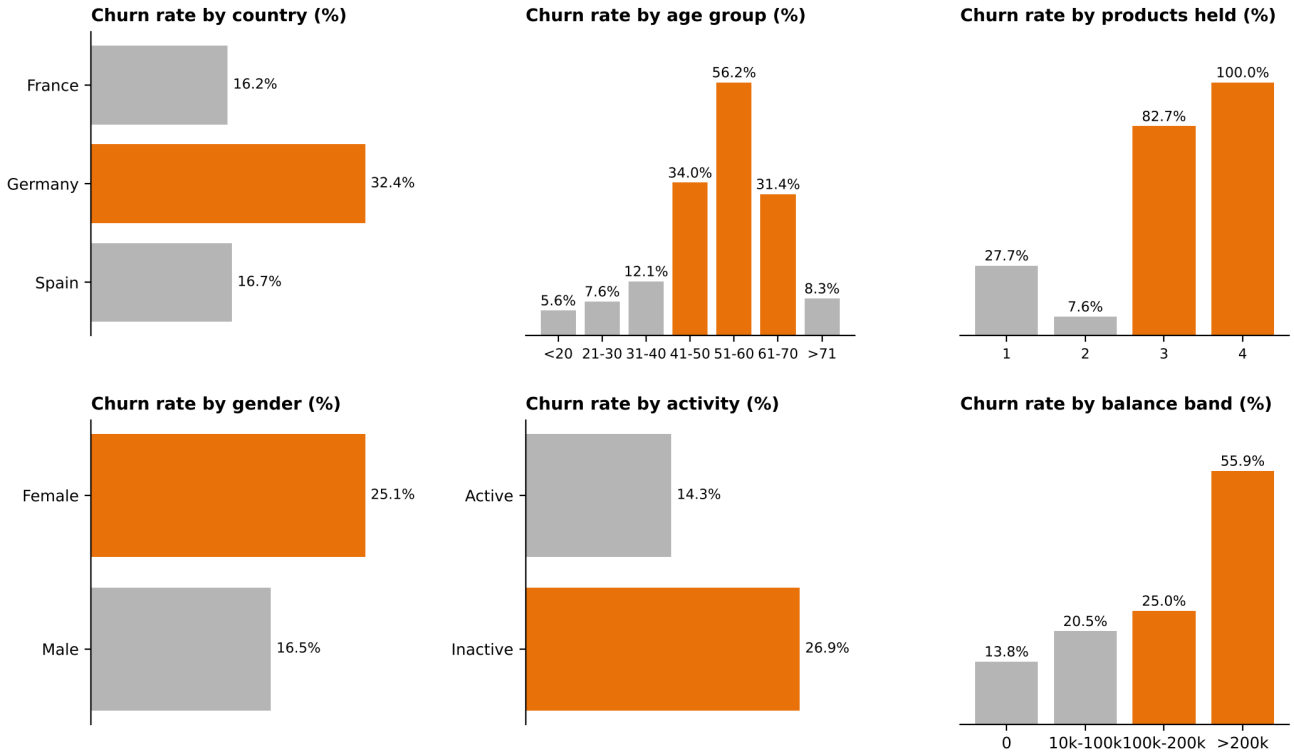
Measures

Measure	Definition
Total Customers	Count of Customer ID
Customers Lost	Count of Churn Status where Churn Status = "Churned"
Churn Rate	Customers Lost divided by Total Customers
Churned Customers	Total Customers where Churn Status = 1 (see technical notes)
Target / Min / Max Churn Rate	15% target, with 0% and 100% bounds, used by the gauge on the Dashboard

Report pages

Page	What it shows
Dashboard	KPI cards (Total Customers, Customers Lost, Churn Rate), gauge against the 15% target, churn by country, gender and age group, a donut of churned vs retained, and slicers for country (with flags), gender, activity status and credit card status.
Report 1	Customer mix donuts (gender, activity, card, country, products), customers and churn rate by credit score band and by balance band, and a scatter of age against balance split by churn status.
Report 2	Churn rate by credit score, tenure and products, a Customers At Risk table, and a ribbon chart of age groups by country.

Findings



Orange bars mark the segments with the highest churn. Rates are the share of customers in each group who churned.

Profile of a churned customer

Measure	Churned	Retained
Average age	44.8	37.4
Average balance	91,109	72,745
Average credit score	645	652
Average tenure	4.9	5.0
Inactive members	63.9%	44.5%
Female customers	55.9%	42.7%

Highest-risk segments

Segment	Customers	Churn rate
Customers holding 3-4 products	326	85.9%
German customers aged 51-60	243	69.5%
Inactive customers in Germany	1,261	41.1%
Female customers in Germany	1,193	37.6%

Small bands should be read with care: balance over 200k has 34 customers, balance 1k-10k has 1, and credit score 400 or below has 19 customers (all churned).

Recommendations and next steps

Recommended actions

Action	Why
Review the German market	Germany has 25% of customers but 40% of churn. Investigate pricing, service and competitor offers there first.
Focus on customers aged 41-60	This group churns at 34% to 56%. Assign relationship managers and offer products suited to this life stage.
Rethink 3-4 product bundles	Customers with three or more products leave at 86%. Review how these bundles are sold and supported.
Re-engage inactive members	Inactive customers churn at 26.9%, nearly twice active ones. Trigger outreach when activity drops.

The data shows where customers leave, not why. Each action should begin with a short investigation before a retention programme is designed.

Technical notes and next steps for the report

- **Churned Customers measure:** it filters on Churn Status = 1, but the column stores the text "Churned", so the measure will not match any rows. Point it at "Churned", or reuse Customers Lost.
- **Churn Rate measure:** wrap the division in DIVIDE() so an empty filter selection returns blank instead of an error.
- **Target, min and max churn rate** are stored as columns repeated on every row. Convert them to measures.
- **Duplicate columns:** Age Group, Credit Scores and Acc. Balance exist in both Customer Data and the banding tables. Use the banding tables in visuals so the sort order works, and hide the duplicates.
- **Add analysis visuals:** Key Influencers and Decomposition Tree for automatic churn drivers, plus drill-through from a country or age band to the customer list.

Conclusion

Churn stands at 20.4% against a 15% target. The report points to four places to act first: Germany, the 41-60 age group, customers with three or more products, and inactive members.